AJMAL ABBAS
Machine Learning Engineer | Generative AI | LLM | Agentic AI
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Portfolio Website: https://skillful-lilac-ec7.notion.site/AJMAL-A-3a4ba46fe55a4b93908e876a9920e435

Professional Summary:

Machine Learning Engineer and AI Researcher with expertise in designing, training, evaluating, and deploying AI/ML solutions across marine ecosystem modeling, cybersecurity, computer vision, and trustworthy multimodal AI. Proven experience developing end-to-end machine learning pipelines, surrogate modeling systems, deep learning architectures, and predictive analytics solutions from research to deployment. Published researcher with IEEE and DSN contributions, including work on confidence-aware AI systems and multimodal learning. Skilled in Python, PyTorch, TensorFlow, LLM-powered applications, model optimization, evaluation, and scalable data-driven solutions. Passionate about applying AI to solve complex real-world challenges across diverse domains.

Skills and Certificates

- **Data Science & AI**: Pytorch, TensorFlow, SciKit-Learn, Machine Learning, Deep Learning, Stats Techniques, NLP, Big-Data, Hugging Face
- **Programming and Tools**: Python, R, SQL, PySpark, Tableau, Django, Minitab, ARC GIS, GeoPandas, Lang flow
- **Domains**: Cybersecurity, Agriculture, Marine Ecosystem, Material Modeling & Business
- **Certifications**: Snowflakes, AI Automation and Agentic AI, **Azure Certificates**: Synapse Analytics, Data Bricks, OpenAI, Data Engineering

Education:

Master of Science in Data Science with 4.0/4.0 CGPA — **Sep 2024-May 2026**
University of Massachusetts Dartmouth, MA, USA

- **Master's Thesis: Emulating Marine Ecosystem Dynamics: A Machine Learning Approach for Biomass Prediction and Forecasting**

Post Graduate Program in Data Science with 3.7/4.0 CGPA — **Sep 2023-May 2024**
Vellore Institute of Technology (VIT), Bangalore, India

- **Project Worked and Published Title:** A **Cutting-Edge Deep Learning Models for Paddy Leaf Disease Detection and Classification**

Published in IEEE Publication 3rd IEEE International Conference on Artificial Intelligence for Internet of Things (ALLOT) (Acceptance Rate 8%)

Work Experience:

Graduate Research Assistant - School of Marine Science and Technology (SMAST)/UMASS Dartmouth, MA, USA — **Sep 2025 – May 2026**

- Formulated and implemented a hybrid ML-DL ecosystem emulator (Random Forest + LSTM) to predict and forecast biomass for 51 marine species using 58 years of NEUS Atlantis simulations, achieving 94% overall accuracy on highly zero-inflated, heterogenous ecological data.
- Engineered a high-performance surrogate modeling pipeline with optuna- optimized Random Forest (0.89s inference) and LSTM forecasting (0.26s inference), enabling near-instant scenario evaluation and long-horizon forecasts to 2035, dramatically reducing computational cost versus full ecosystem simulations.

Graduate Research Assistant – College of Engineering /UMASS Dartmouth, MA, USA — **Jun 2025- Aug 2025**

- Architected and created a hybrid software vulnerability detection framework combining AST stylometry features with CodeBERTa-small-v1 embeddings, enabling scalable evaluation across Devign, MVD and ReVeal datasets and achieving 80% classification accuracy and packaging the system as a reusable python library(hybrid-vul) for practical vulnerability analysis.
- Streamlined end to end preprocessing and training pipelines by benchmarking CPU vs GPU feature extraction and demonstrating faster MLP training using CPU-based features (2.61 vs 7.76s), enabling efficient deployment on low-power, resource -constrained systems.

Graduate Research Assistant- College of Engineering /UMASS Dartmouth, MA, USA — **Dec 2024 - Jan 2025**

- Implemented and prepared a deep learning-based Network Intrusion Detection (NIDS) leveraging Graph Neural Network (GNNs) to model complex network traffic, detect anomalies and identify cyber threats in real time, achieving 97% detection accuracy.
- Integrated AI-driven security strategies using state-of the art GNN and deep learning models, improving threats detection accuracy and system scalability in dynamic network environments through close collaboration with peers and faculty.

Project Experience:

From Weak Embeddings to Trustworthy Multimodality: Modality-Agnostic Learning with Confidence-Aware Fallback (Published in SIEDS 2026)
Sep 2025 - Dec 2025

- Deployed unified multimodal learning pipeline (text-image-video) using contrastive embedding, integrating CIFAR-100 ResNet, CLIP, BLIP captioning and R3D-18 video models to produce calibrated confidence scores across modalities.
- Improved model trustworthiness through probability calibration (ECE, temperature scaling), lowering calibration error and supporting confidence- driven multimodal decision fallback.

SmartLeafNet: Hybrid Deep Learning and Machine Learning Model for Leaf Disease Detection — **Jan 2025 - May 2025**

- Constructed and integrated SmartLeafNet, a model integrating Deep Learning techniques for accurate leaf disease across 22000 augmented images.
- Accomplished accuracy 70% and macro F1-score with a lightweight, deployment- ready system suitable for real- time inference on edge devices.

DeepFake Detection — **Sep 2024- Dec 2024**

- Innovated a deep learning framework combining CNNs, LSTMs and attention mechanisms for accurate detection of deepfake videos using celeb-DF dataset and achieved 90% of accuracy.
- Delivered robust deepfake detection performance, enabling reliable identification of fraudulent video content and enhancing media authenticity verification through governance and data integrity strategies

Foliage Doctor: Recognizes Leaf Disease and Recommends Treatments to Restores Leaf Health — **Jan 2024 - May 2024**

- Launch of Detectron2, a revolutionary algorithm for the identification of leaf diseases.
- Developed a high-performance algorithm achieving 5x faster runtime than models by enabling rapid and accurate.

Instagram User Analytics — **Jan 2024 - May 2024**

- Analyzed Instagram user activity and posting trends using SQL to identify peak engagement periods.
- Generate Insights to optimize product launch timing and marketing campaigns.

Publications:
VulStyle: A Multi-Modal Pre-Training for Code Stylometry-Augmented Vulnerability Detection (DSN 2026 Acceptance rate 19%)